



EU Legal Frameworks and Discrimination Challenges in AI-Enhanced Criminal Justice Systems

RIMC09: EU Criminal Policy and Justice in Context

Ainhoa Zamora Martínez

Student number: 692678

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Supervisor: Dr. Abdessamad Bouabid

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1. Introduction

The new regulatory environment which, examines the EU's attempt to build a "*common criminal justice area*", is too optimistic. This goal is complicated by the deep tension between investments in AI for Law Enforcement Agencies (LEAs) and the push for the protection of fundamental rights. For this reason, academics are now drawing on Beck's risk society approach, specifically Curran (2018), who diagnosed a lack of political consideration and public debate regarding the risks of digital economies in general and the rise of artificial intelligence, in particular. Also, Kusche (2024) argues that the situation has now changed, with the European legislation aiming to regulate digital services, digital markets, and the application of artificial intelligence (AI Act) in order to prevent harm and not leave socio-technological trajectories to the profit-oriented decisions of digital platforms and corporations.

At the core of this tension lies the data itself. Contemporary societies are marked by structural inequalities, including discrimination based on race, gender, and migration status. Because AI technologies "learn" from historical data, these systems frequently reproduce and amplify existing social biases (Mohanty & Vyas, 2018). This has led many voices to warn about the capacity of AI to impact negatively on a variety of human rights. (Aizenberg & van den Hoven, 2020). As Persak (2018) argues in her examination of EU criminal law legitimization, any EU action in criminal matters must be grounded in substantive principles that go beyond mere effectiveness, which appears *prima facie* sufficient but ultimately falls short in technological solutions.

This paper examines: **Are the current EU legal frameworks equipped to manage the non-discrimination challenges introduced by machine learning models and AI in the criminal justice system?**

This paper contributes to debates in EU Justice and Home Affairs by examining how internal security policies are operationalised through predictive technologies. Although the empirical analysis relies on advanced machine-learning models, its core relevance lies in assessing how data-driven tools are embedded within EU criminal justice cooperation and internal security governance, where preventive risk assessment has become a central policy instrument.

It also argues that these frameworks, though essential, are inadequate in addressing practical empirical shortcomings. Using the quantitative results from a predictive crime model for Barcelona as a case study, the analysis tests a "high-risk" AI system under EU standards.

The structure of this paper is as follows: it first outlines the theoretical and legal context of AI use in current regulations and directives, it also explains the methodological approach and empirical case study structures, before analysing the regulatory implications of the findings and concluding with a critical assessment of their scientific and societal importance.

2. Methodology

This paper adopts a criminological risk-governance lens, combining EU criminal law legitimisation theory with empirical AI risk analysis. A case study methodology aiming to evaluate the effectiveness of EU regulatory frameworks in governing AI applications in criminal justice. The use of case studies is ideal for bridging abstract legal norms with operational realities, allowing for an in-depth examination of how regulations perform in practice (Yin, 2018). The approach triangulates legal-doctrinal analysis—interpreting EU laws like the AI Act¹, GDPR² and LED³ with empirical quantitative data from a predictive crime model developed for Barcelona.

The empirical component draws from machine learning techniques, specifically Random Forest and XGBoost⁴. This setup simulates a "high-risk" AI system under the AI Act (Art.6), enabling assessment of regulatory requirements for robustness (Art.15) and risk management (Art.9). Further, the empirical analysis follows three steps: (1) training RF and XGBoost models on historical crime data; (2) testing robustness

¹ Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence

² Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data.

³ Directive (EU) 2016/680 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data by competent authorities for the purposes of the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, and on the free movement of such data.

⁴ These models were trained on geospatial crime data from 2017-2024, incorporating variables like temporal patterns, land use classifications, and socioeconomic indicators.

through cross-validation and structural break analysis; and (3) evaluating the results against EU legal requirements on accuracy, robustness, and non-discrimination.

The methodology relates to the research question by testing whether regulatory safeguards meaningfully mitigate bias and ensure reliability under dynamic conditions. In policing contexts, predictive inaccuracies are not neutral technical errors: they redistribute suspicion and enforcement unevenly across space and population, with amplified consequences for areas marked by migration and socioeconomic marginalisation (Schuilenburg & Soudijn, 2023; Wonders, 2017).

While this methodology offers tangible evidence of regulatory gaps, as opposed to purely theoretical critiques, it also has limitations. First, the Barcelona dataset reflects enforcement priorities—such as patrol density—rather than objective crime distribution, potentially amplifying biased feedback loops (Richardson et al., 2019). Second, findings lack generalisability due to Barcelona’s unique demographics and the significant variance in domestic administrative capacity and political priorities across EU Member States.

3. Predictive Methods as a “High-Risk AI System”

Predictive algorithms in law enforcement, like those for crime forecasting, pose significant risks to fundamental rights, including privacy, non-discrimination, and the presumption of innocence. The U.S. Case of *State v Loomis* (2017)⁵ exemplifies these challenges: the COMPAS⁶ recidivism tool was upheld despite unverifiable biases, setting a precedent for regulatory hurdles that the EU seeks to overcome. In the EU context, these tools (under the AI Act) are classified as “high-risk” due to their potential to exacerbate inequalities, particularly in migration and border control, where AI-driven surveillance intersects with *crimmigration* practices. While US jurisprudence does not bind EU institutions, this case has informed EU policy debates; the EC’s (2020)

⁵ *State v Loomis* (2017), where Eric Loomis pleaded guilty to fleeing police officers in a vehicle in 2013. He was sentenced to 6 years imprisonment and 5 years under supervision. The court argued that the judge treated COMPAS as one factor among many, not determinative. And that there was no evidence showing that COMPAS was inaccurate in this specific case. This case, and the subsequent discovery of its racial bias, crystallised the core challenges for regulators: transparency, bias, and due process.

⁶ This AI technology named COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) employed to determine the convicts’ risk of reoffending when deciding penalties.

research on AI argued that algorithmic flaws are justifications for the risk-based regulatory approach in the AI Act.

3.1. Data protection and Privacy Frameworks

The EU's first line of defence against algorithmic harms was its data protection framework, yet it remains fragmented and insufficient for law enforcement. Kinchin (2024) frames GDPR transparency as implying a 'right to explanation' with consent as central. However, this framing obscures critical deficiencies. There is a bifurcation between the GDPR for general processing and the LED for policing which creates opportunities for regulatory arbitrage, allowing authorities to exploit distinctions and expand powers (Vermeulen, 2007).

To contextualise, place-based predictive models like the Barcelona case might not process "personal data" if focused on geographic hotspots, so will evade protections despite disproportionate impacts on migrant communities facing heightened surveillance (Franko Aas & Gundhus, 2015). But the consent paradigm is illusory in coercive policing environments, where individuals—often migrants or refugees—lack autonomy. This category error transposes market-oriented privacy models to state and police power, undermining rights and perpetuating crimmigration dynamics (Aliverti, 2012). Thus, while the framework addresses some risks, it fails to prevent violations of evidence presumptions in predictive identification (Richardson et al., 2019).

These deficiencies appear within the JHA context, where data processing is closely linked to migration control and the policing of mobility. Considering a hypothetical but possible scenario: a predictive risk-assessment tool deployed by FRONTEX to analyse mobility data from Schengen border crossings. Because the system targets “irregular migration flows” rather than named individuals, operators may classify it as non-personal data processing. Yet the downstream effect (that certain nationalities have intensified checks at specific borders), produces individualised consequences that GDPR was designed to prevent.

3.2. The AI Act

The EU AI Act has been widely celebrated as a global regulatory milestone. However, closer examinations reveal an instrument heavily shaped by industry and law

enforcement interests, often at the expense of fundamental rights. As Access Now (2025) argues, the Act is “an unambitious piece of product safety legislation” rather than a human rights framework, “full of loopholes, carve-outs, and exceptions”.

The regulation adopts a risk-based approach, designating law enforcement AI systems as “high-risk” and prohibiting certain AI practices under Art. 5. Recital 42 specifically cites concerns about predictive identification “labelling individuals potential criminals before they have engaged in any criminal conduct” (Levano, 2024).

First, the “high-risk filter” loophole Art.6(3) allows developers to self-assess that their systems do not pose significant risks, effectively opting out of regulatory obligations. However, FRA (2025) warns that this filter could introduce “loopholes” and limit the protection of fundamental rights in place. Second, the national security exemption, Art.2(3) is so broad that member states can invoke it to bypass core protection entirely. And as EDRi (2024) notes, this “creates a substantial risk of abuse” given the blurry line between national security and other activities by law enforcement agencies—authorities could subject protests to AI surveillance by citing national security and other activities by law enforcement agencies.

While the AI act establishes *de jure* safeguards through risk classification, documentation, and accuracy requirements for predictive policing, the extent to which these guarantees translate into *de facto* protection against discriminatory outcomes in everyday practices remains underexamined. Moreover, some academic literature on data-driven policing consistently highlights risks of discrimination, bureaucratisation, and self-reinforcing surveillance patterns, particularly in the over-policing of minority neighbourhoods (Schuilenburg & Soudijn, 2023). From a rule-of-law perspective, scholars also emphasise concerns related to dirty data, function creep, and violations of the presumption of innocence (Richardson et al., 2019). These risks are especially acute in predictive identification practices, which invert the logic of criminal law by attributing suspicion before any offence has occurred—a problem explicitly acknowledged in Recital 42 of the AI Act (Levano, 2024). However, despite this extensive legal critique, there remains a lack of empirical, quantitative evidence demonstrating how such risks materialise in real-world European policing contexts. Regulatory frameworks discuss “risk” in the abstract, but what does this risk look like in

practice? Do these “high-risk” models actually deliver the efficiency that supposedly justifies their use?

3.3. Case study of EU Policy Risk

The Barcelona predictive model serves as an empirical test using longitudinal crime data from 2017 to 2024. Classified as a "high-risk" system under the AI Act (Art.6), the model exemplifies how machine learning failures in robustness and accuracy can perpetuate discriminatory outcomes, particularly in diverse urban settings like Barcelona, where migration patterns intersect with crime hotspots (Wonders, 2017). By analysing the model's performance metrics, this case study tests the practical efficacy of EU requirements for technical robustness (AI Act, Art.15), risk management (Art.9), and data quality (LED, Art.4), revealing gaps that undermine non-discrimination protections. The table below compares the model’s performance when trained on all data with data excluding the pandemic disruption.

Table 1: Cross-Validation Performance by Test Year

	Test Year	XGBOOST		RF	
		MAE	R²	MAE	R²
Full cross-validation	2017	35.8	0.429	13.7	0.949
	2018	28.7	0.863	25.3	0.880
	2020	26.5	0.603	41.4	-0.326
	2021	17.8	0.782	26.8	-0.306
	2022	17.8	0.940	18.2	0.916
	2023	30.0	0.644	20.0	0.874
	2024	12.1	0.949	8.3	0.983
	Mean±SD	23.2±8.1	0.771±0.195	21.9±9.9	0.614±0.575
No-Covid validation	2017	35.8	0.429	13.7	0.949
	2018	28.7	0.863	25.3	0.880
	2019	17.2	0.959	21.4	0.945
	2022	16.1	0.898	13.0	0.947
	2023	21.9	0.773	8.5	0.991
	2024	20.5	0.803	9.1	0.986
	Mean±SD	23.4±7.5	0.787±0.188	15.2±6.8	0.950±0.040

Commented [AZ1]: modificar texto e indicar que es del 2017 al 2024

Random Forest predictions were worse than random guessing during those years; rather than identifying the structural mechanisms of crime, the algorithm had internalised a fixed pattern of pre-pandemic activity—a sign of overfitting rather than learning. This demonstrates that the model cannot generalise beyond the narrow social and temporal scope of its training data, which violates the accuracy and robustness principles central to both the AI Act (art.15) and LED (art.4).

The model's COVID-era collapse could alternatively indicate sensitivity to genuine structural change in crime drivers. However, the EU AI Act requires high-risk systems to remain reliable under unforeseen conditions, so what may be informative for researchers constitutes a robustness failure under EU law.

Crucially, this also challenges the regulatory assumption that “technical validation” equates to “trustworthiness”. Under the AI Act’s risk-based framework, a model can be certified as “high-performing” based on historical cross-validation yet still fail catastrophically when confronted with real-world instability. This paradox—a model that performs perfectly in theory but fails in practice—mirrors the core critique advanced by scholars like Aizenberg & van den Hoven (2020): that algorithmic legality is being defined by performance metrics, not by lived social outcomes. The results obtained, thus offer empirical evidence of this gap between formal compliance and substantive justice.

From a JHA perspective, the instability observed in both RF and XGB models raises fundamental questions about the policy rationale for predictive policing within EU internal security strategies. JHA governance increasingly assumes that technological tools enhance efficiency in policing and migration control. However, as this case study demonstrates, predictive models may fail under conditions of social disruption. Similar critiques have been raised in the context of border risk analysis, where automated assessments often exacerbate uncertainty while masking political responsibility behind technical systems (Franko Aas & Gundhus, 2015).

4. Conclusion

This paper has examined whether the current EU legal frameworks are equipped to address the challenges introduced by ML models and AI systems. Using a

criminological risk-governance lens and a case study of predictive policing in Barcelona, the analysis demonstrates that, while the EU has constructed a sophisticated regulatory architecture, a substantial gap remains between formal legal compliance and substantive protection.

At the legal level, EU frameworks formally recognise the risk posed by predictive technologies and seek to mitigate them through accuracy, robustness, and risk management. However, the analysis demonstrated that the safeguards are fragmented and vulnerable to regulatory evasion. The current bifurcation between the GDPR and the LED, combined with self-assessment mechanisms from the AI Act, enables predictive practices while limiting meaningful accountability.

Barcelona's case study provides empirical evidence of these shortcomings. The failure of predictive performance—evidenced by the model's negative R^2 values—is not merely a technical outlier but a proof of algorithmic fragility, and the struggle of these models to generalise beyond social and temporal conditions of their training data.

From a scientific perspective, this paper contributes to existing literature by moving beyond abstract critiques of algorithmic risk and providing quantitative evidence of how predictive policing systems may fail in real-world European contexts. The societal and policy implications are particularly salient within the Justice and Home Affairs domain. As this case study shows, algorithmic instability may exacerbate rather than mitigate uncertainty, redistributing suspicion and enforcement in ways that disproportionately affect already marginalised communities.

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